Paul Nguyen

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EDUCATION

Boston University, *College of Engineering & Kilachand Honors College*, Boston, MA — Expected May 2028
B.S. in Mechanical Engineering and Electrical Engineering | *GPA*: 3.50/4.00
Relevant Coursework: Signals and Systems, Thermodynamics, Fluid Mechanics, Manufacturing, Electromagnetics

EXPERIENCE

Structures/Avionics Team Member, *BU Rocket Propulsion Group*, Boston, MA — Nov 2025 – Present

- Designed and optimized 4+ structural components using CAD and FEA, cutting overall structural mass by 15% while meeting all load requirements for team's liquid propulsion test stand.
- Integrated structural designs with propulsion and systems teams (30+ members), managing 4 test-stand assembly cycles and 3 cold-flow/hot-fire tests with zero structural failures.
- Identified and fixed 6 design weaknesses through structural analysis, saving an estimated 40+ hours of rework.

Hackathon Co-Organizer, *Boston University UAV Team,* Boston, MA — Sep 2026 – Present

- Coordinated monthly UAV hackathons for 40+ students across 8–10 teams, managing a $1,500 semester budget, venue logistics, judging, and 3 mentor/sponsor relationships.
- Grew hackathon participation by 24 students (150%) in 2 months by launching 3 hands-on technical workshops (flight controllers, CAD, ROS) for students new to UAV development.

IT Support Specialist - LETS, *BU Information Services & Technology*, Boston, MA — Jan 2025 – Present

- Delivered on-site AV/IT support for 100+ Registrar classrooms across 2 campuses, resolving 30–40 support tickets per week.
- Troubleshoot Windows/macOS systems, classroom AV, and Crestron control systems, resolving 90% of issues on first visit with an average response time under 10 minutes.

PROJECTS

DUM-E Robotic Arm Replica, Personal Project — May 2026 – Present

- Built a multi-axis robotic arm replica of DUM-E from the Iron Man films, designing, 3D printing, and assembling 30+ custom parts over 150+ hours and 3 design revisions.
- Developed a voice-control system in Python, mapping 10+ spoken commands to coordinated arm movements, writing 1,000+ lines of code and testing across 4 hardware/software iterations.

Battery Degradation in MBTA Public Transport eVehicles, Researcher — Jan 2026 – May 2026

- Collaborated with 4 mechanical engineering students to analyze how fast charging and low-temperature cycling degrade NMC 111 lithium-ion batteries in MBTA electric bus fleets, synthesizing 25+ peer-reviewed studies.
- Investigated 3 degradation mechanisms (lithium-ion diffusion, intergranular cracking, and SEI layer growth) and proposed 2 materials engineering solutions (single-crystal cathodes and dopant incorporation) to improve battery durability and lifespan.

HTS Junction Box Design & Integration, BU Rocket Propulsion Group — Nov 2025 – Mar 2026

- Designed and documented HTS electrical junction box with a 6-person multidisciplinary team, integrating 20+ sensors and actuators (pressure transducers, thermocouples, solenoid valves) across liquid propulsion test stand.
- Reduced wiring complexity by consolidating 14 connectors into 6, standardizing pin assignments across 40+ channels, and writing full pinout and wiring documentation, cutting assembly time by ~30%.

SKILLS

Programming: MATLAB, Simulink, Python (NumPy, Pandas), C, C++, Java, JavaScript, G-Code, UNIX, Git
CAD & Manufacturing: SolidWorks, Fusion 360, Onshape, FEA, GD&T, CAM, DFM, 3D Printing, CNC Machining, Machine Shop
Electronics: KiCad, LTspice, Altium, PCB Design, Soldering, Microcontrollers, Oscilloscopes, Wire Harnessing, Power Electronics, Sensor Integration, DAQ, Motor Control, Arduino, Raspberry Pi, PID Control, I2C Protocols
IT & Software: Windows/macOS, Crestron, Classroom AV, Microsoft Office, Google Workspace, VS Code, Adobe Creative Cloud
Languages: English (Native), Vietnamese (Native)